

EXPT.
NO.

NAME

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YOUVA

- hand shield must be worn
- Process should be done in ventilation area
- Set voltage current etc properly.

Questions:-

Q1) What is the working principle of Arc Welding?

Arc welding uses a coated consumable electrode. This coating produces a gas shield and slag to protect the weld from atmosphere. To start the arc, welder briefly touches the tip of the electrode to the work piece to complete the circuit then the electrode is removed to a slight distance from the work piece, this initiates an arc. The arc so produced generates intense heat which melts the tip of the electrode, the coating and a portion of the adjacent base metal. This molten metal is allowed to solidify together to form a solid joint.

Q2) How do you select the polarity in case of DC welding?

Two types of polarities are used for DC welding

(i) DCSA (Direct Current Straight Polarity)

→ When work is positive and electrode is negative

(ii) DCRP (Direct Current Reverse Polarity)

→ When work is negative and electrode is positive for welding this works as the heat generated is split into two parts in ratio of 66% at the positive pole and 33% at the negative pole.

Teacher's Signature: _____